

Week 3 Lab ENGN1218

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PRELAB QUESTIONS

0.1. Theoretical Calculation

Bands 1	brown
Bands 2	green
Bands 3	black
Bands 4	brown
Bands 5	brown
R (ohms)	$1.5k\Omega \pm 1\%$

Bands 1	orange
Bands 2	orange
Bands 3	black
Bands 4	brown
Bands 5	brown
R (ohms)	$3.3k\Omega \pm 1\%$

Bands 1	green
Bands 2	blue
Bands 3	black
Bands 4	brown
Bands 5	brown

R (ohms)	$5.6k\Omega \pm 1\%$
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Bands 1	grey
Bands 2	red
Bands 3	black
Bands 4	brown
Bands 5	brown
R (ohms)	$8.2k\Omega \pm 1\%$

Bands 1	brown
Bands 2	black
Bands 3	black
Bands 4	orange
Bands 5	brown
R (ohms)	$100k\Omega \pm 1\%$

1. Simple Resistive Circuit

1.1. Theoretical Calculation

$$V = IR$$

$$I = \frac{V}{R}$$

$$\text{Given } R = R_1 = 8.2k$$

$$V = V_1 = 8.2k$$

$$I = \frac{10}{8.2 * 10^3}$$

$$= 1.2 * 10^{-3} \Omega \text{ The other one is } = 8.3 \times 10^{-3}$$

2. Simple Resistive Circuit

2.1. Theoretical Calculation

$$V = IR$$

$$I_{\text{total}} = \frac{V_{\text{total}}}{R_{\text{total}}}$$

$$\text{Given } R = R_1 = 100k$$

$$V = V_1 + V_2 = 10 + 5V = 15V$$

$$I = \frac{15}{100 * 10^3}$$

$$= 150\mu\Omega$$

$$V_x = V_{\text{total}} = 15V$$

	Voltage $V_x(V)$	Resistance R_1 (k Ω)	Current I(mA)
Theoretical	15	100	0.15
Simulated	15	100	0.15
Measured			

3. DC Voltage Sources with Opposite Polarities

3.1. Theoretical Calculation

Using KVL. lets go clockwise around the center.

$$V_2 - V_1 - IR_1 = 0$$

$$I = \frac{V_2 - V_1}{R_1}$$

$$I = \frac{10 - 5}{100 * 10^{-3}}$$

$$I = \frac{10 - 5}{100 * 10^{-3}}$$

$$I = 50\mu A$$

$$V_x = I * R_1$$

$$V_x = 50 \times 10^{-6} * 100 * 10^3$$

$$V_x = 5$$

	Voltage V_x (V)	Resistance R_1 (k Ω)	Current I(mA)
Theoretical	5	100	0.05
Simulated	5	100	0.05
Measured			

4. Simple Resistive Circuit

4.1. Theoretical Calculation

Bands 1	Voltage drop across resistor (V)			Current through resistor (mA)		
grey	R_1	R_2	R_3	R_1	R_2	R_3
Theoretical	2.0971	2.9029	2.9029	0.0014	0.0009	0.0005
Simulated	2.1	2.9	2.9	0.0014	0.00088	0.00052
Measured						

5. Resistor Ladder Circuit

5.1. Theoretical Calculation